



BRAC University
Department of Mathematics and Natural Sciences
Assignment-1
Semester: Summer -26
Course Title: Integral Calculus and Differential Equations
Course Code: MAT 120

Deadline: -

Total Marks: 10

Answer all the questions:

1. Using Riemann sum find the area under the curve $f(x) = x^2 - 7x + 14$ on the interval $[0, \frac{7}{2}]$ with x_k^* as (i) left end point (ii) middle point and (iii) right end point. [2]
2. Using Riemann sum find the area under the curve $f(x) = x^3 - 2x^2 + 4$ on the interval $[1, 4]$ with x_k^* as (i) left end point (ii) middle point and (iii) right end point for $n = 6$. [2]
3. Evaluate the following integrals: [2]
 - i) $\int \left(3x \cos(4 - x^2) - \frac{8x\sqrt{4-x^2}}{(2-x)^{\frac{3}{2}}(2+x)^{\frac{3}{2}}} \right) dx$
 - ii) $\int \frac{x^5 + x^4 + 4x^3 + 4x^2 + 4x + 4}{(x^2 + 2)^3} dx$
 - iii) $\int \ln(x + \sqrt{x^2 + a^2}) dx$
4. Derive reduction formula for the following integrals: [1]
 - (i) $\int x^n \sin x dx$
 - (ii) $\int \tan^n x dx$
5. Evaluate in terms of beta functions: $\int_0^1 \left(\frac{x^2}{\sqrt{1-x^5}} \right) dx$ [2]
6. Given: [1]

$$f(x) = \begin{cases} 9 - 2e^x, & \text{if } x > 0, \\ 8 \sin x, & \text{if } x \leq 0 \end{cases}$$

Find

$$\int_{-2}^4 f(x) dx.$$